

Centre Number	Candidate Number	Candidate Name
---------------	------------------	----------------

NAMIBIA SENIOR SECONDARY CERTIFICATE

BIOLOGY ORDINARY LEVEL

6116/2

PAPER 2

1 hour 30 minutes

Marks 80

2024

Additional Materials: Ruler
Non - programmable calculator

INSTRUCTIONS AND INFORMATION TO CANDIDATES

- Candidates answer on the Question Paper in the spaces provided.
- Write your Centre Number, Candidate Number and Name in the spaces at the top of this page.
- Write in dark blue or black pen.
- You may use a soft pencil for any diagrams, graphs or rough working.
- Do not use correction fluid.
- You may use a non-programmable calculator.
- Do not write in the margin *For Examiner's Use*.
- Answer **all** questions.
- The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's use	
1	
2	
3	
4	
5	
6	
7	
8	

Total	
-------	--

Marker	
Checker	

This document consists of **15** printed pages and **1** blank page.



Republic of Namibia
MINISTRY OF EDUCATION, ARTS AND CULTURE

- 1 Fig.1.1 shows a housefly.

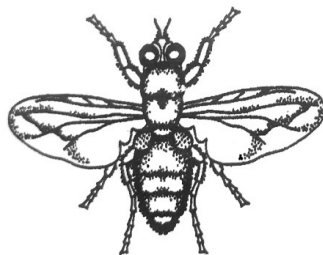


Fig. 1.1

- (a) *Musca domestica* is the scientific name given to a housefly.

Name the system for naming organisms.

..... [1]

- (b) Describe **four** features visible in Fig. 1.1 that identifies the housefly as an insect.

.....

[4]

- (c) Insects are classified as arthropods.

- (i) Name **two** other classes of arthropods.

1
 2

[2]

- (ii) Outline **three** ways that scientists make use of the hierarchical classification system.

1
 2
 3

[3]

[10]

- 2 Fig. 2.1 shows the human alimentary canal and its associated parts.

For
Examiner's
Use

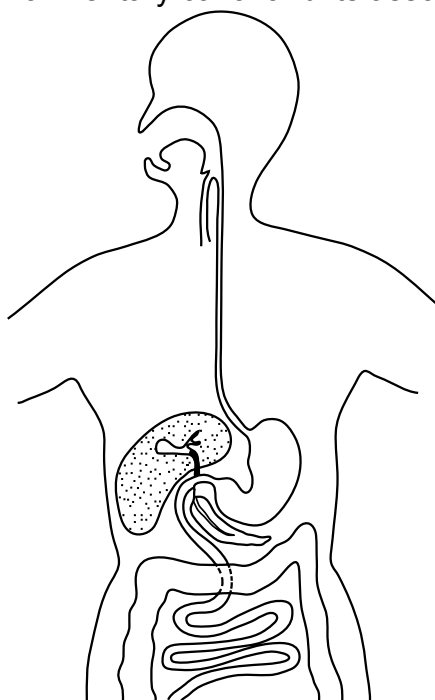


Fig. 2.1

- (a) Define the terms *ingestion* and *absorption*.

ingestion.....
.....
absorption
.....

[2]

- (b) On Fig. 2.1, using **label lines** and **letters**, indicate with letter

P, the part where digestion of proteins begins,

B, the site where bile is stored,

L, the part where enzyme lipase is produced,

W, the part where the absorption of water is completed.

[4]

- (c) The process of chemical digestion is carried out by protein molecules called enzymes.

- (i) Name the basic units that make up enzyme molecules.

..... [1]

- (ii) Name the final product of starch digestion.

..... [1]

(iii) Distinguish between mechanical and chemical digestion.

.....

.....

.....

.....

.....

[2]

[10]

- 3 (a) Describe **two** similarities between the lymphatic and circulatory systems.

1

.....

2

.....

[2]

- (b) Below is a list of different levels of organisation.

Arrange the levels of organisation, starting from the largest.

organ, organism, cell, organ system, tissue

largest

.....

.....

.....

smallest.....

[1]

- (c) Fig. 3.1 shows blood components. The small fragments are responsible for blood clotting.

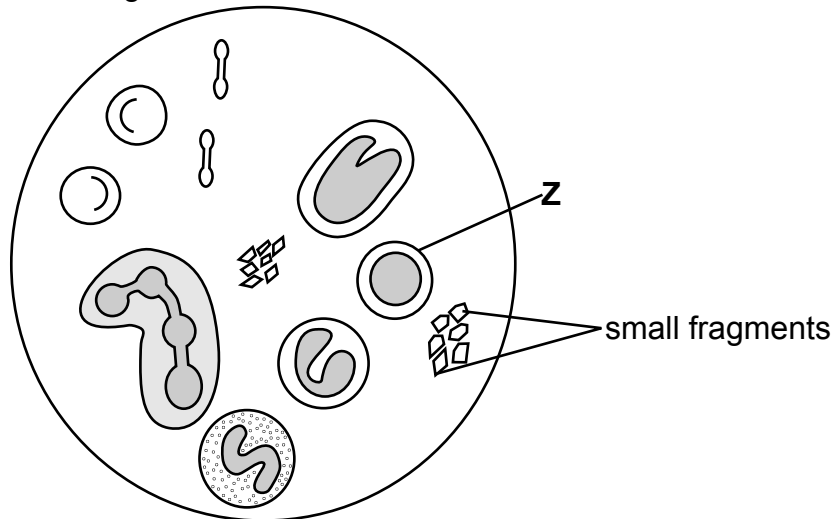


Fig. 3.1

- (i) Identify component **Z**.

[1]

- (ii) Describe the process of blood clotting.

[4]

- (d) Blood is transported to the heart through the veins.

Discuss **two** ways in which veins are structurally adapted to this function.

1

.....

2

.....

[2]

[10]

- 4 (a) Distinguish between *breathing* and *respiration*.

.....

.....

.....

.....

.....

[2]

- (b) Fig. 4.1 shows the effects of exercise on the breathing pattern.

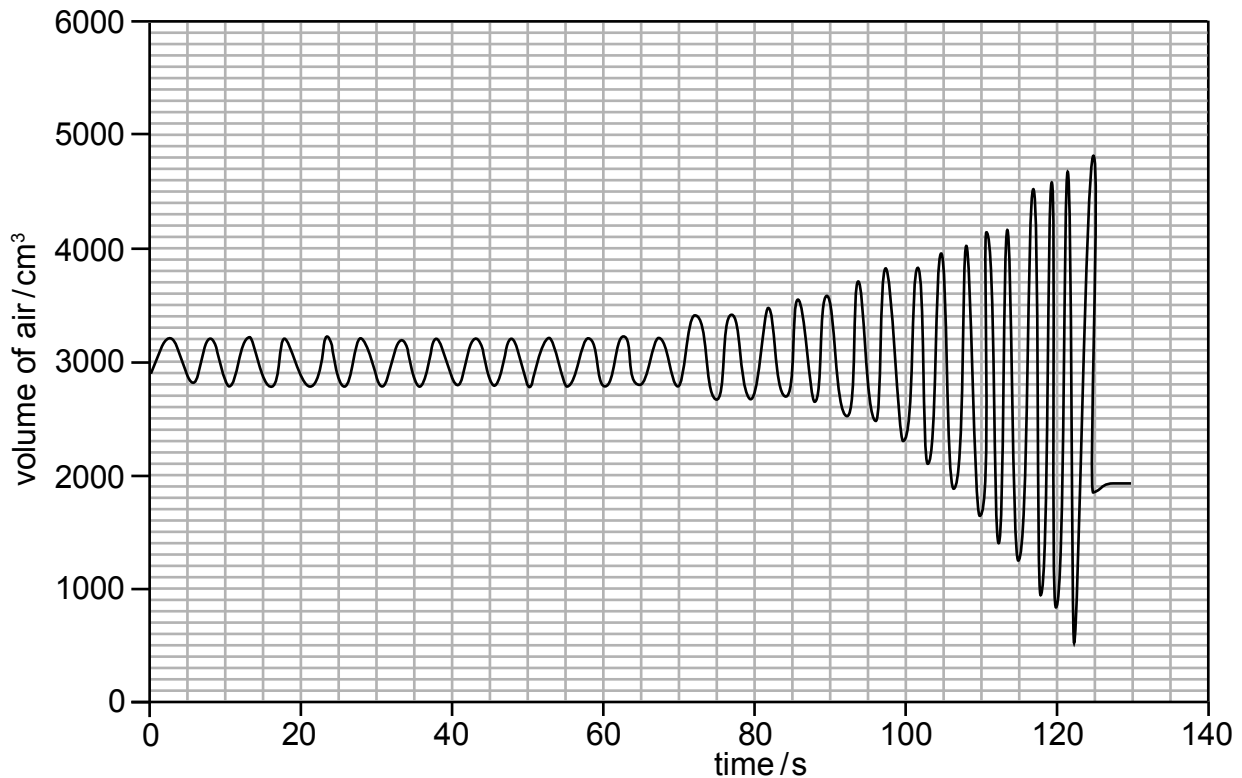


Fig. 4.1

One breath is counted as inhalation and exhalation. The mean volume of air breathed in per breath at rest is 500 cm³.

- (i) Using information from the graph, calculate the volume of air breathed in per minute at rest.

[2]

- (ii) After 70 seconds, the person started exercising.

Describe and explain the changes in the breathing pattern.

.....

.....

.....

.....

.....

.....

.....

.....

[4]

- (c) Describe the effects of carbon monoxide on the gas exchange system.

.....

.....

.....

.....

[2]

[10]

- 5 Fig. 5.1 summarises the process of photosynthesis in a chloroplast.

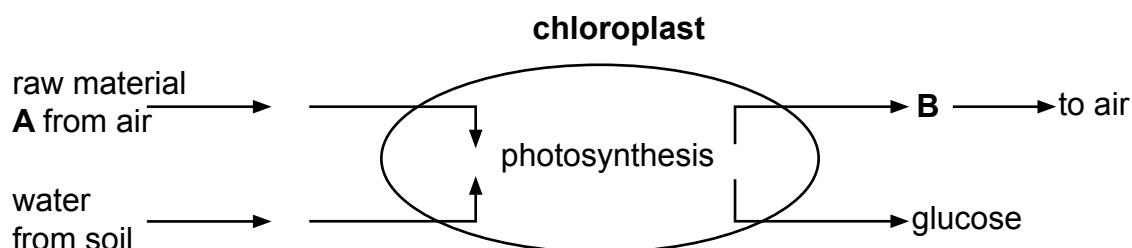


Fig. 5.1

- (a) Name raw material **A** and substance **B** in Fig. 5.1.

raw material **A**

substance **B**

[2]

- (b) Photosynthesis produces glucose which can be converted into different substances.

Complete Fig. 5.2 by filling in the missing information at (i), (ii) and (iii) to show some of the uses of glucose in plants.

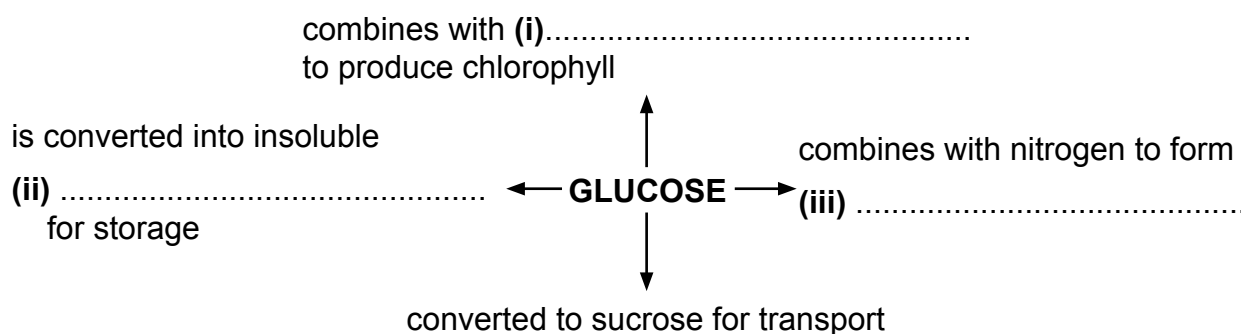


Fig. 5.2

[3]

(c) Fig. 5.3 shows a cross-section of a dicotyledonous stem.

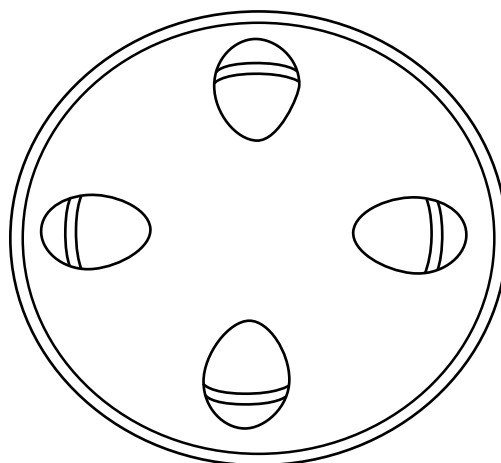


Fig. 5.3

Describe how translocation of sucrose can be used to control pests.

.....

.....

.....

.....

.....

.....

[3]

(d) Explain the mechanism by which water moves up the plant, until it leaves through the stomata.

.....

.....

.....

.....

.....

.....

[3]

[11]

- 6 Fig. 6.1 shows a cell dividing by meiosis.

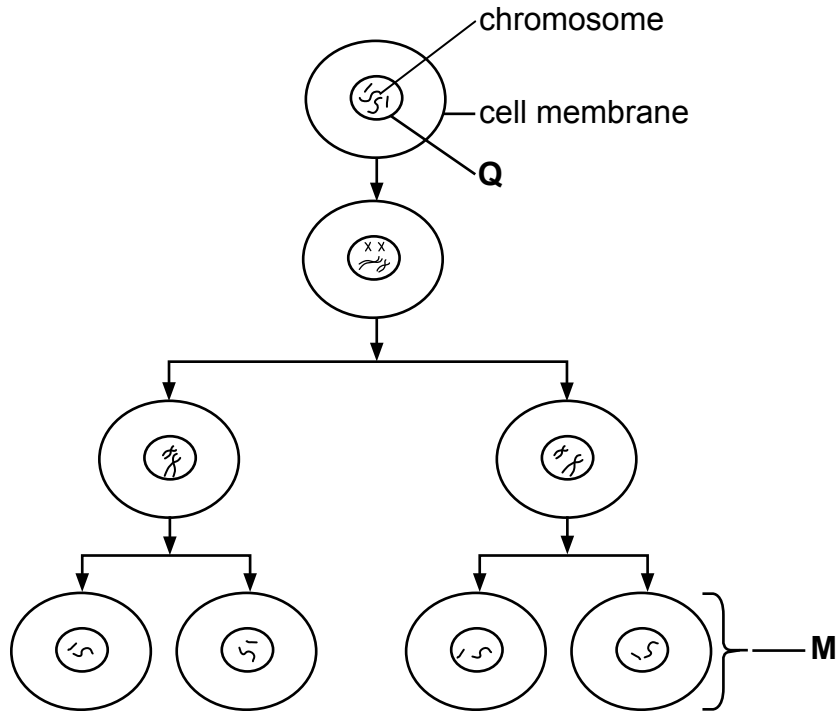


Fig. 6.1

- (a) Identify the cell structure labelled **Q**.

..... [1]

- (b) State the diploid number of chromosomes in this cell.

..... [1]

- (c) Name the organ in male animals in which meiosis occurs.

..... [1]

- (d) Give the specific name of cells labelled **M**, if this dividing cell is from the female animal.

..... [1]

- (e) With reference to Fig. 6.1, identify **two** features that show that this cell division is meiosis.

1

.....

2

.....

[2]

(f) Another type of cell division is mitosis.

Outline **three** roles of mitosis in living organisms.

1

.....

2

.....

3

.....

[3]

[9]

- 7 Fig. 7.1 shows an incomplete life cycle of a flowering plant.

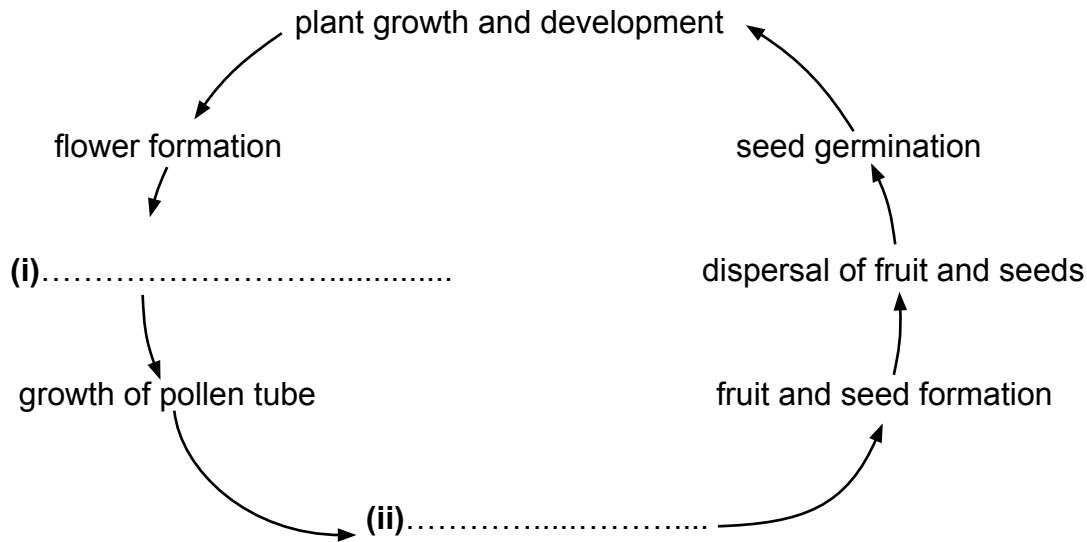


Fig. 7.1

- (a) Complete Fig. 7.1 by filling in the processes missing at (i) and (ii). [2]
- (b) Explain how the following environmental conditions affect seed germination.

oxygen

.....

water

.....

[2]

- (c) Give **two** reasons why seed dispersal is important.

1

.....

2

.....

[2]

- (d) Apart from the dry mass method, state **two** other methods of measuring growth and discuss one advantage for each.

method 1

advantage

.....

method 2

advantage

.....

[4]

[10]

- 8 Match the following terms with the correct descriptions using ruled lines. One match has been done for you. Each description may only be used once or not at all.

term	description
(a) (i) evolution	the observable features of an organism
(ii) homozygous	differences between individuals of the same species
(iii) phenotype	length of DNA that codes for a protein
(iv) recessive	a process of change over a period of time
(v) variation	having two identical alleles of a particular gene
	an allele that is only expressed when there is no dominant allele

[4]

- (b) Variation can result in conditions such as sickle-cell anaemia.

key: HH - normal haemoglobin, this individual does not have sickle-cell anaemia

Hh – this individual does not have sickle-cell anaemia, but can pass the sickle-cell anaemia gene to offspring

hh – this individual has sickle-cell anaemia

- (i) Outline the symptoms of sickle-cell anaemia.

.....

.....

.....

.....

[2]

- (ii) Use the alleles given in the key, to explain with a genetic diagram how a child can inherit sickle-cell anaemia when both parents do not suffer from this condition.

[4]

[10]

BLANK PAGE

The DNEA acknowledges the usage and reproduction of third party copyright material in the NSSC Assessment, **with and without permission** from the copyright holder. The Namibian Government Copyright Act allows copyright material to be used limitedly and fairly for educational and non-commercial purposes.

The Directorate of National Examinations and Assessment operates under the auspices of the Ministry of Education, Arts and Culture in Namibia.